

Rebuttal Figures and Tables

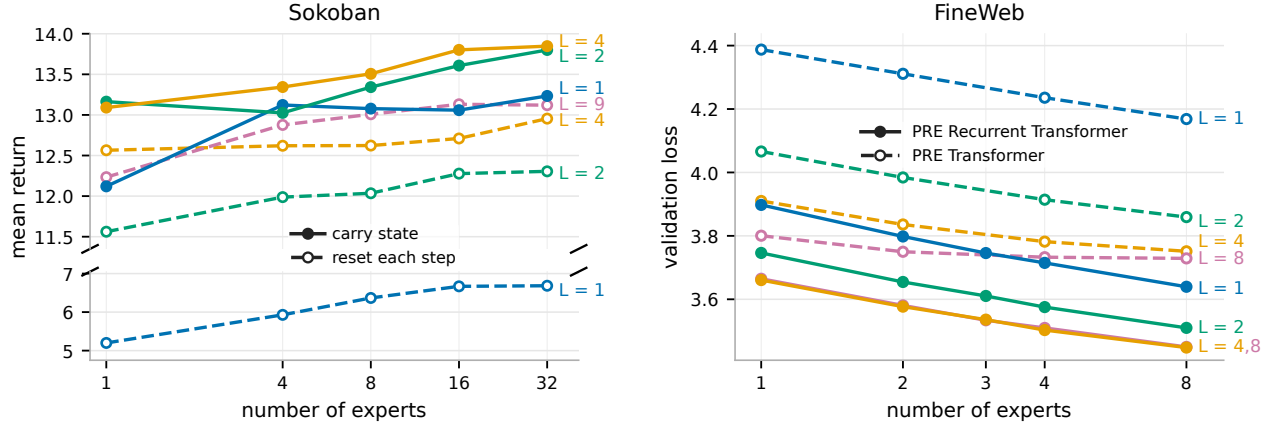


Figure 2: Carried state supplies a serial computation axis. **Left:** In Sokoban, solid lines carry hidden state across environment steps and dashed lines reset it every step. Resetting state severely hurts shallow updates, while extra within-step depth partially recovers performance; color denotes within-step depth L . **Right:** In FineWeb next-token prediction, the PRE Recurrent Transformer carries latent recurrent state across token steps, while the PRE Transformer recreates latent state from the current segment at each step. The context window size is 128 for both architectures.

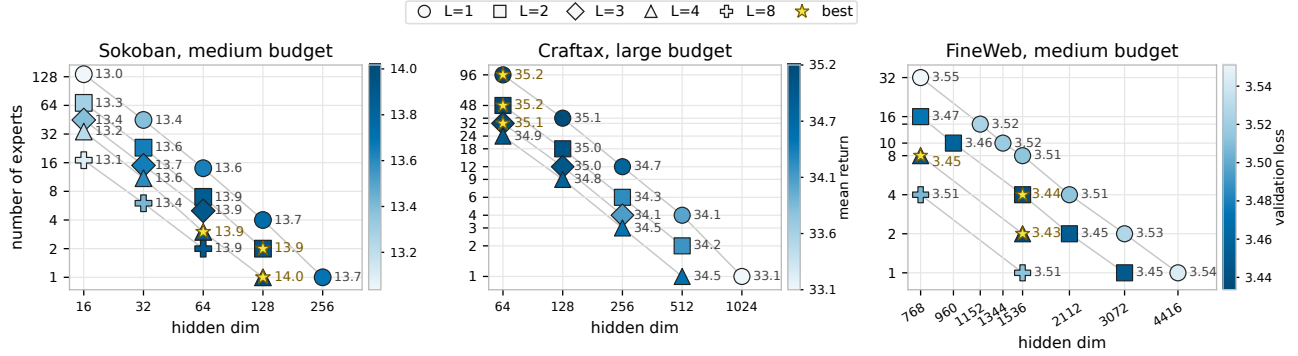


Figure 4: Fixed-compute allocation across domains. The horizontal axis varies expert hidden dimension, the vertical axis varies experts per depth level, and marker shape denotes within-step depth.

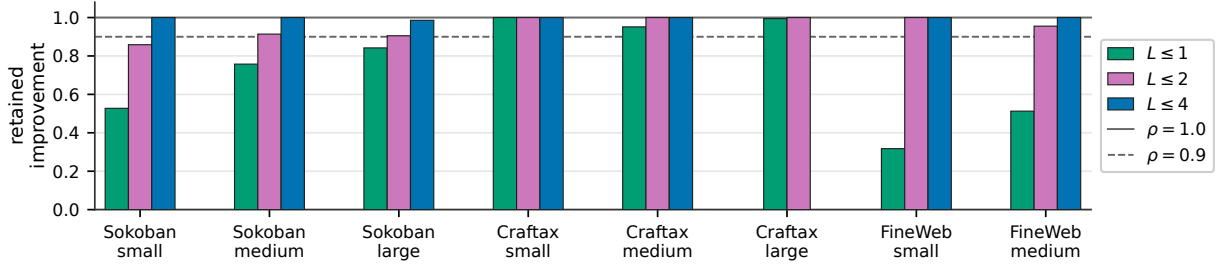


Figure 5: Retained improvement under different depth caps. Each bar reports the retained-improvement ratio $\rho_{\leq k}$ for one matched-budget sweep under depth caps $k \in \{1, 2, 4, 8\}$, with Q defined as episodic return for Sokoban and Craftax and negative validation cross-entropy for FineWeb. Horizontal reference lines mark $\rho = 1.00$, which matches the unrestricted best configuration, and $\rho = 0.90$, which retains 90% of the best improvement over the domain reference.

Table 1: Best observed performance under within-step depth restrictions in the fixed-compute sweeps. Sokoban and Craftax report episodic return, where higher is better. FineWeb next-token prediction reports validation loss, where lower is better. Gap reports the performance loss from the unrestricted optimum: return drop for Sokoban and Craftax, and validation-loss increase for FineWeb. Max L is the largest within-step depth included in that sweep.

Domain	Budget	Max L	Best overall	Best with $L \leq 2$	Gap	Best with $L \leq 1$	Gap
Sokoban	small	8	13.665	13.552	0.113	13.287	0.378
Sokoban	medium	8	14.017	13.918	0.099	13.738	0.279
Sokoban	large	8	14.228	14.099	0.129	14.013	0.215
Craftax	small	8	32.436	32.436	0.000	32.436	0.000
Craftax	medium	8	33.934	33.934	0.000	33.802	0.132
Craftax	large	4	35.200	35.200	0.000	35.179	0.021
FineWeb	small	8	3.481	3.481	0.000	3.559	0.078
FineWeb	medium	8	3.434	3.441	0.007	3.511	0.078

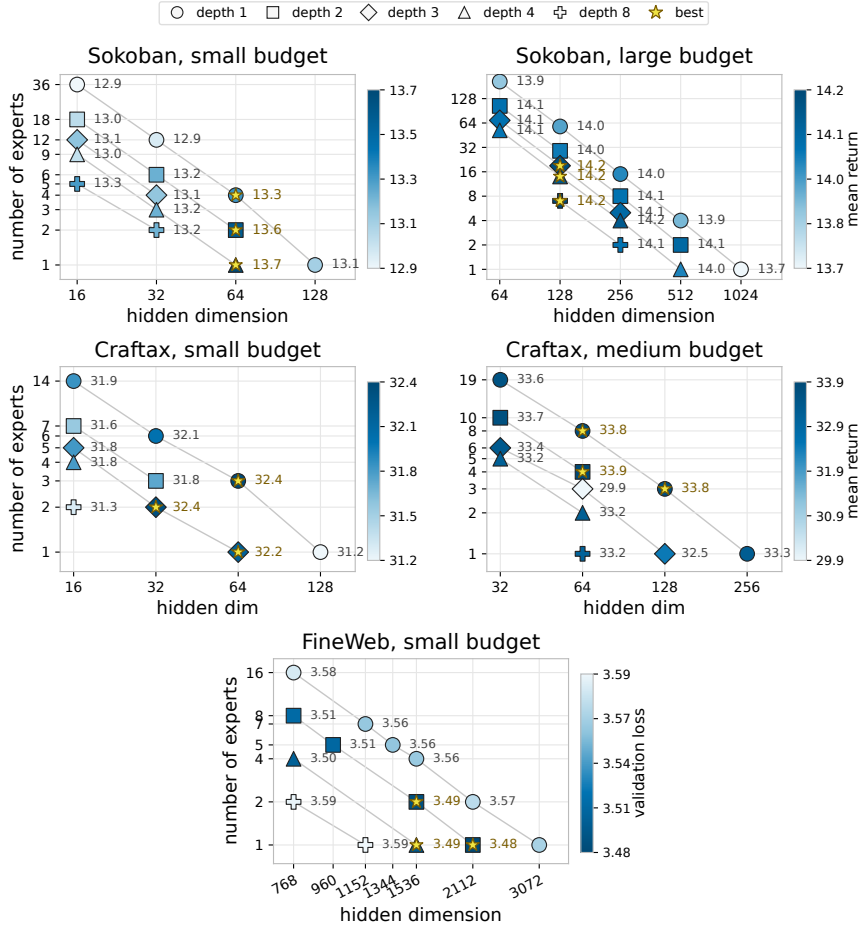


Figure 9: Additional fixed-compute allocation budgets. **Top left:** Sokoban small budget. **Top right:** Sokoban large budget. **Middle left:** Craftax small budget. **Middle right:** Craftax medium budget. **Bottom:** FineWeb small budget. In Sokoban and Craftax, panels vary expert hidden dimension, experts per within-step depth level, and within-step depth; color shows mean return, and only completed or near-completed configurations are shown. In FineWeb, color shows validation loss at the common comparison point of 1.30B consumed tokens, where lower is better.

Table 2: Fixed-compute budgets. Relative work is normalized to the smallest budget within each domain. For Sokoban, $C(d) \propto d(d + 16)$; for Craftax, $C(d) \propto d(256 + 2d)$. Best reports the best observed performance in the matched-budget sweep: return for Sokoban and Craftax, and validation loss for FineWeb.

Domain	Budget	Reference config	Work / small budget	Best
Sokoban	small	$L = 1, E = 1, d = 128$	1.00	13.665
Sokoban	medium	$L = 1, E = 1, d = 256$	3.78	14.017
Sokoban	large	$L = 1, E = 1, d = 1024$	57.78	14.228
Craftax	small	$L = 1, E = 1, d = 128$	1.00	32.436
Craftax	medium	$L = 1, E = 1, d = 256$	3.00	33.934
Craftax	large	$L = 1, E = 1, d = 1024$	36.00	35.200
FineWeb	small	$L = 16, E = 1, d = 768$	1.00	3.481
FineWeb	medium	$L = 32, E = 1, d = 768$	2.00	3.434

Table 3: Best exact-depth comparison, $L = 4$ versus $L = 2$. For each depth, the configuration with the best observed mean higher-is-better score Q is selected. ΔQ is the observed difference between sample means $\bar{Q}_4 - \bar{Q}_2$, so positive values favor $L = 4$. The p -value is from Welch’s t -test of $H_0 : \mathbb{E}[Q_4] \leq \mathbb{E}[Q_2]$ against $H_1 : \mathbb{E}[Q_4] > \mathbb{E}[Q_2]$. n_4/n_2 gives the numbers of independent runs for the two configurations.

Domain	Budget	Best $L = 4$	Best $L = 2$	ΔQ	n_4/n_2	p
Sokoban	small	$E = 1, d = 64$	$E = 2, d = 64$	0.113	9/9	0.0150
Sokoban	medium	$E = 1, d = 128$	$E = 2, d = 128$	0.099	9/9	0.0060
Sokoban	large	$E = 14, d = 128$	$E = 2, d = 512$	0.074	3/3	0.1005
Craftax	small	$E = 4, d = 16$	$E = 3, d = 32$	0.063	3/3	0.4407
Craftax	medium	$E = 2, d = 64$	$E = 4, d = 64$	-0.690	3/3	0.9275
Craftax	large	$E = 24, d = 64$	$E = 48, d = 64$	-0.266	3/3	0.8364

Table 4: Best exact-depth comparison, $L = 2$ versus $L = 1$. For each depth, the configuration with the best observed mean higher-is-better score Q is selected. ΔQ is the observed difference between sample means $\bar{Q}_2 - \bar{Q}_1$, so positive values favor $L = 2$. The p -value is from Welch’s t -test of $H_0 : \mathbb{E}[Q_2] \leq \mathbb{E}[Q_1]$ against $H_1 : \mathbb{E}[Q_2] > \mathbb{E}[Q_1]$. n_2/n_1 gives the numbers of independent runs for the two configurations.

Domain	Budget	Best $L = 2$	Best $L = 1$	ΔQ	n_2/n_1	p
Sokoban	small	$E = 2, d = 64$	$E = 4, d = 64$	0.265	9/9	< 0.0001
Sokoban	medium	$E = 2, d = 128$	$E = 4, d = 128$	0.180	9/9	< 0.0001
Sokoban	large	$E = 2, d = 512$	$E = 15, d = 256$	0.086	3/3	0.0745
Craftax	small	$E = 3, d = 32$	$E = 3, d = 64$	-0.670	3/3	0.8968
Craftax	medium	$E = 4, d = 64$	$E = 8, d = 64$	0.132	3/3	0.3551
Craftax	large	$E = 48, d = 64$	$E = 96, d = 64$	0.021	3/3	0.4545

Table 5: Best $L = 4$ versus all $L = 2$ configurations in Sokoban. For each small and medium budget, the $L = 4$ configuration with the best observed mean return is compared with every $L = 2$ configuration. Positive ΔQ favors $L = 4$, and p is from the one-sided Welch test defined above. n_4/n_2 gives the numbers of independent runs for the $L = 4$ and $L = 2$ configurations.

Budget	Best $L = 4$	$L = 2$ configuration	ΔQ	n_4/n_2	p
small	$E = 1, d = 64$	$E = 2, d = 64$	0.113	9/9	0.0150
small	$E = 1, d = 64$	$E = 6, d = 32$	0.468	9/9	< 0.0001
small	$E = 1, d = 64$	$E = 18, d = 16$	0.658	9/9	< 0.0001
medium	$E = 1, d = 128$	$E = 2, d = 128$	0.099	9/9	0.0060
medium	$E = 1, d = 128$	$E = 7, d = 64$	0.151	9/9	< 0.0001
medium	$E = 1, d = 128$	$E = 23, d = 32$	0.375	9/9	< 0.0001
medium	$E = 1, d = 128$	$E = 68, d = 16$	0.730	9/9	< 0.0001

Table 6: Best $L = 2$ versus all $L = 1$ configurations in Sokoban. For each small and medium budget, the $L = 2$ configuration with the best observed mean return is compared with every $L = 1$ configuration. Positive ΔQ favors $L = 2$, and p is from the one-sided Welch test defined above. n_2/n_1 gives the numbers of independent runs for the $L = 2$ and $L = 1$ configurations.

Budget	Best $L = 2$	$L = 1$ configuration	ΔQ	n_2/n_1	p
small	$E = 2, d = 64$	$E = 4, d = 64$	0.265	9/9	< 0.0001
small	$E = 2, d = 64$	$E = 1, d = 128$	0.481	9/7	0.0002
small	$E = 2, d = 64$	$E = 12, d = 32$	0.624	9/9	< 0.0001
small	$E = 2, d = 64$	$E = 36, d = 16$	0.688	9/9	< 0.0001
medium	$E = 2, d = 128$	$E = 4, d = 128$	0.180	9/9	< 0.0001
medium	$E = 2, d = 128$	$E = 1, d = 256$	0.231	9/9	< 0.0001
medium	$E = 2, d = 128$	$E = 14, d = 64$	0.277	9/9	< 0.0001
medium	$E = 2, d = 128$	$E = 45, d = 32$	0.534	9/9	< 0.0001
medium	$E = 2, d = 128$	$E = 136, d = 16$	0.881	9/9	< 0.0001

The Sokoban small- and medium-budget rows contain nine runs, except for the small-budget $L = 1, E = 1, d = 128$ anchor, which contains seven. Seeds 7 and 9 for that anchor ended below the established 94.2% training-progress cutoff and are excluded. Large-budget Sokoban and Craftax use the available three runs.

Table 7: Per-seed Sokoban small- and medium-budget fixed-compute results. These are all points shown for this benchmark in Figures 4 and 9. Each seed entry is the mean of the final five logged episodic returns. Mean and standard error are computed across the available seed values. Columns $s_1, \dots, s_9$ identify random seeds; a dash denotes an unavailable seed or an undefined single-run standard error.

Budget	L	E	d	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9	Mean	SE
small	1	1	128	12.874	13.109	13.269	12.726	13.146	13.063	–	13.313	–	13.071	0.079
small	1	4	64	13.381	13.174	13.499	13.164	13.195	13.196	13.309	13.332	13.334	13.287	0.038
small	1	12	32	12.755	12.926	12.909	13.052	13.096	12.947	13.003	12.787	12.875	12.928	0.038
small	1	36	16	12.796	13.082	13.020	13.203	12.969	12.945	12.702	12.517	12.549	12.865	0.079
small	2	2	64	13.620	13.700	13.518	13.643	13.345	13.553	13.536	13.423	13.631	13.552	0.038
small	2	6	32	13.223	13.185	13.296	13.080	13.178	13.017	13.121	13.380	13.297	13.198	0.038
small	2	18	16	13.172	13.124	13.022	13.214	12.515	12.964	12.836	13.289	12.931	13.008	0.078
small	3	4	32	12.963	13.223	13.195	13.007	12.995	13.141	13.146	13.005	13.407	13.120	0.048
small	3	12	16	13.155	13.485	13.082	13.177	13.175	13.161	12.883	13.023	12.981	13.125	0.056
small	4	1	64	13.559	13.757	13.660	13.775	13.689	13.605	13.687	13.530	13.727	13.665	0.029
small	4	3	32	13.415	13.370	13.130	13.087	13.092	13.013	13.098	13.257	13.301	13.196	0.048
small	4	9	16	12.703	13.142	12.948	12.890	12.860	13.133	12.966	13.098	13.161	12.989	0.052
small	8	2	32	13.273	13.252	13.324	13.221	13.124	13.238	13.150	12.905	13.211	13.189	0.041
small	8	5	16	13.516	13.325	12.831	13.521	13.296	13.071	13.108	13.183	13.514	13.263	0.079
medium	1	1	256	13.713	13.635	13.750	13.737	13.803	13.593	13.703	13.596	13.654	13.687	0.024
medium	1	4	128	13.672	13.722	13.860	13.715	13.805	13.753	13.756	13.742	13.613	13.738	0.024
medium	1	14	64	13.674	13.652	13.646	13.730	13.596	13.667	13.631	13.478	13.696	13.641	0.024
medium	1	45	32	13.422	13.281	13.384	13.482	13.186	13.315	13.443	13.372	13.564	13.383	0.038
medium	1	136	16	13.097	13.130	13.101	12.942	13.072	13.035	12.949	12.849	13.158	13.037	0.034
medium	2	2	128	13.862	13.868	13.986	13.845	13.823	13.962	13.881	14.068	13.965	13.918	0.027
medium	2	7	64	13.873	13.821	13.890	13.917	13.893	13.777	13.795	13.896	13.935	13.866	0.019
medium	2	23	32	13.656	13.576	13.784	13.670	13.594	13.541	13.713	13.668	13.577	13.642	0.026
medium	2	68	16	13.349	13.436	13.109	13.278	13.443	13.475	13.214	13.085	13.198	13.287	0.049
medium	3	5	64	13.905	13.899	13.838	14.056	13.783	13.885	13.884	13.913	13.932	13.899	0.025
medium	3	15	32	13.781	13.524	13.604	13.781	13.684	13.385	13.475	13.833	13.806	13.653	0.054
medium	3	45	16	13.150	13.451	13.194	13.537	13.449	13.466	13.450	13.417	13.350	13.385	0.044
medium	4	1	128	13.970	14.079	13.974	14.074	13.958	14.025	13.914	14.109	14.050	14.017	0.022
medium	4	3	64	13.833	14.091	13.875	13.761	13.736	14.068	13.967	14.129	13.802	13.918	0.050
medium	4	11	32	13.624	13.596	13.726	13.587	13.428	13.578	13.631	13.632	13.504	13.590	0.028
medium	4	34	16	13.039	13.434	13.324	13.162	13.107	13.230	13.178	13.295	13.211	13.220	0.040
medium	8	2	64	14.012	13.900	13.884	13.718	13.864	13.869	14.036	13.658	14.102	13.894	0.048
medium	8	6	32	13.644	13.339	13.385	13.601	12.965	13.221	13.366	13.581	13.626	13.414	0.075
medium	8	17	16	13.074	13.384	13.148	13.400	12.863	13.178	12.924	12.843	13.128	13.105	0.068

Table 8: Per-seed Sokoban large-budget fixed-compute results. These are all points shown for this benchmark in Figures 4 and 9. Each seed entry is the mean of the final five logged episodic returns. Mean and standard error are computed across the available seed values. Columns $s_1, \dots, s_3$ identify random seeds; a dash denotes an unavailable seed or an undefined single-run standard error.

Budget	L	E	d	s_1	s_2	s_3	Mean	SE
large	1	1	1024	13.783	13.573	13.790	13.715	0.071
large	1	4	512	13.907	13.862	13.971	13.914	0.032
large	1	15	256	14.004	13.999	14.035	14.013	0.011
large	1	58	128	13.990	13.852	14.027	13.956	0.053
large	1	208	64	13.809	13.931	13.891	13.877	0.036
large	2	2	512	14.051	14.070	14.176	14.099	0.039
large	2	8	256	14.141	13.899	14.148	14.063	0.082
large	2	29	128	14.048	13.974	14.102	14.041	0.037
large	2	104	64	14.063	14.067	14.079	14.070	0.005
large	3	5	256	14.142	14.057	14.090	14.096	0.025
large	3	19	128	14.232	14.168	14.226	14.209	0.020
large	3	69	64	13.988	14.103	14.104	14.065	0.039
large	4	1	512	13.957	14.104	14.007	14.022	0.043
large	4	4	256	14.194	14.089	14.208	14.164	0.037
large	4	14	128	14.125	14.214	14.179	14.173	0.026
large	4	52	64	14.042	14.035	14.081	14.053	0.014
large	8	2	256	14.143	14.039	14.010	14.064	0.040
large	8	7	128	14.145	14.281	14.259	14.228	0.042

Table 9: Per-seed Craftax fixed-compute results. These are all points shown for this benchmark in Figures 4 and 9. Each seed entry is the mean of the final five logged episodic returns. Mean and standard error are computed across the available seed values. Columns $s_1, \dots, s_3$ identify random seeds; a dash denotes an unavailable seed or an undefined single-run standard error.

Budget	L	E	d	s_1	s_2	s_3	Mean	SE
small	1	1	128	29.480	31.632	32.538	31.216	0.907
small	1	3	64	32.367	32.800	32.142	32.436	0.193
small	1	6	32	31.592	32.166	32.393	32.050	0.238
small	1	14	16	31.771	31.964	31.866	31.867	0.056
small	2	3	32	32.132	32.141	31.026	31.766	0.370
small	2	7	16	31.171	31.600	31.977	31.582	0.233
small	3	1	64	32.755	31.168	32.701	32.208	0.520
small	3	2	32	32.198	32.390	32.557	32.381	0.104
small	3	5	16	32.047	31.144	32.337	31.843	0.359
small	4	4	16	31.927	31.769	31.791	31.829	0.049
small	8	2	16	31.191	31.510	31.252	31.318	0.098
medium	1	1	256	33.611	33.255	33.034	33.300	0.168
medium	1	3	128	33.481	33.792	34.011	33.761	0.154
medium	1	8	64	33.573	33.952	33.881	33.802	0.116
medium	1	19	32	33.299	33.933	33.635	33.622	0.183
medium	2	4	64	34.372	34.060	33.370	33.934	0.296
medium	2	10	32	33.556	33.955	33.658	33.723	0.120
medium	3	1	128	33.713	30.396	33.326	32.478	1.047
medium	3	3	64	34.636	21.646	33.337	29.873	4.130
medium	3	6	32	33.660	33.012	33.606	33.426	0.207
medium	4	2	64	32.895	33.153	33.685	33.244	0.233
medium	4	5	32	33.349	32.919	33.252	33.173	0.130
medium	8	1	64	33.488	33.150	32.814	33.150	0.194
large	1	1	1024	33.044	34.154	31.969	33.056	0.631
large	1	4	512	33.405	34.443	34.306	34.051	0.326
large	1	12	256	34.329	34.919	34.959	34.736	0.204
large	1	36	128	35.190	35.049	35.172	35.137	0.044
large	1	96	64	35.228	35.285	35.024	35.179	0.079
large	2	2	512	34.128	34.553	33.895	34.192	0.193
large	2	6	256	34.223	33.614	34.942	34.260	0.384
large	2	18	128	34.646	34.683	35.735	35.021	0.357
large	2	48	64	35.490	35.131	34.980	35.200	0.151
large	3	4	256	33.728	34.438	34.156	34.107	0.206
large	3	12	128	34.878	35.186	34.882	34.982	0.102
large	3	32	64	34.726	35.700	34.995	35.140	0.291
large	4	1	512	34.670	34.525	34.392	34.529	0.080
large	4	3	256	34.273	34.422	34.738	34.477	0.137
large	4	9	128	34.966	34.542	34.787	34.765	0.123
large	4	24	64	35.020	34.583	35.199	34.934	0.183

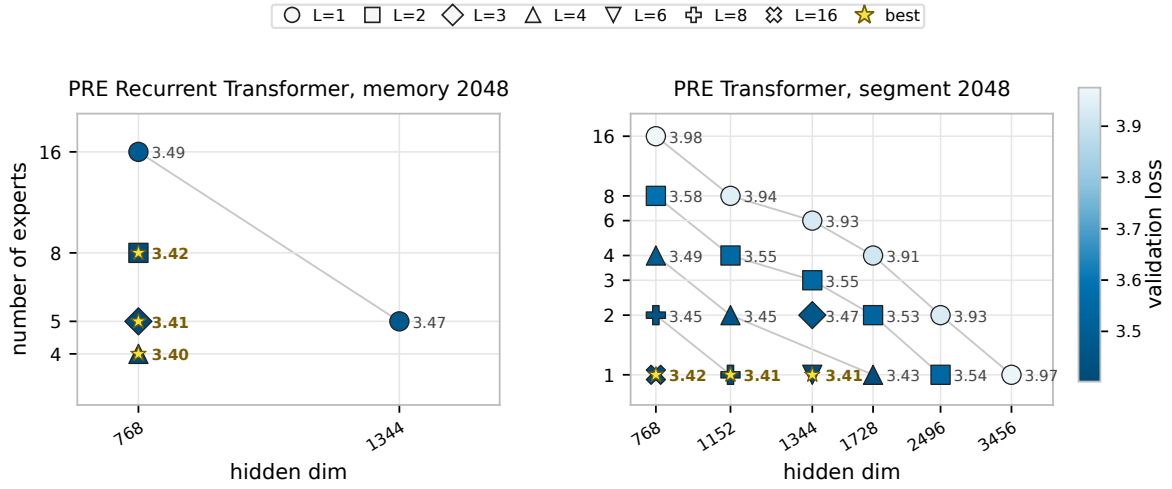


Figure 12: FineWeb budget-16 comparison at long context. **Left:** completed finite PRE Recurrent Transformer runs with recurrent memory length 2048 and segment length of 64. **Right:** the corresponding non-recurrent GPT-style PRE Transformer fixed-budget sweep with training segment length 2048. Color and point labels show validation loss at the latest common logged validation point within each panel, where lower is better; marker shape denotes within-step depth L , and stars mark the best configurations within each panel. The PRE Recurrent Transformer panel contains the subset of memory-2048 configurations that completed with finite validation loss, while the GPT-style PRE Transformer panel contains the completed segment-2048 sweep.

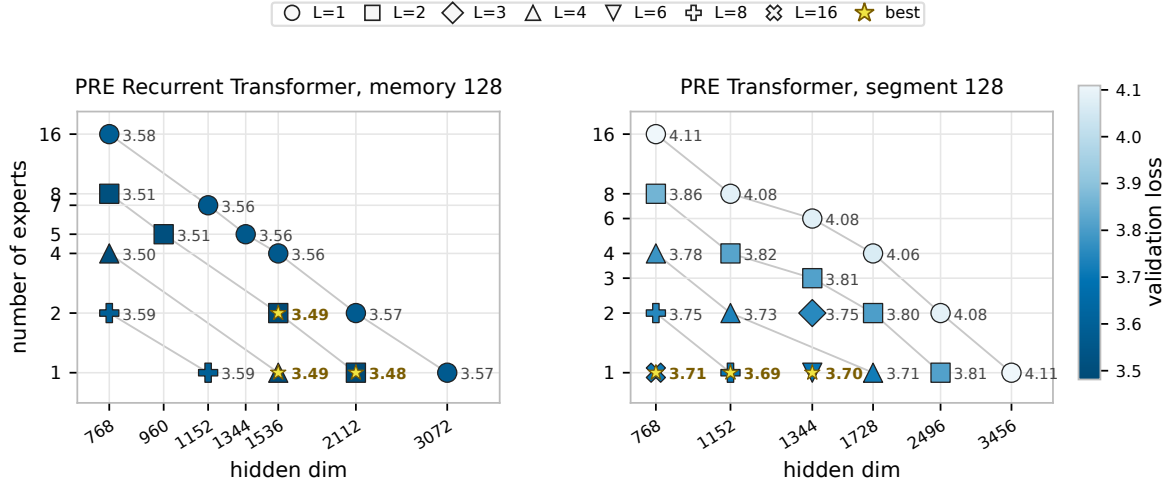
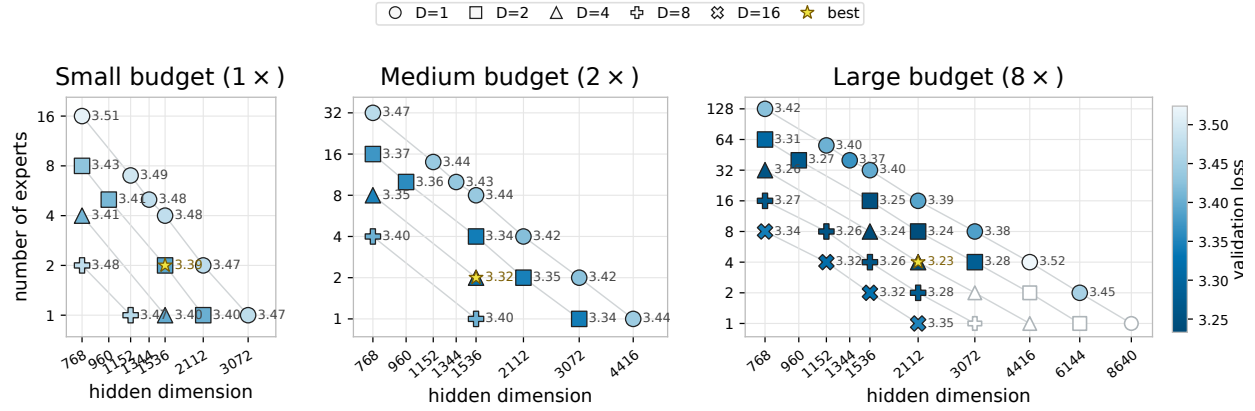


Figure 13: FineWeb budget-16 comparison at context length 128. **Left:** the PRE Recurrent Transformer small-budget allocation map, repeated here for direct comparison. The segment length is 64 and the memory length is 128. **Right:** a non-recurrent GPT-style PRE Transformer sweep with training segment length 128 and the same nominal budget. Color and point labels show validation loss at the latest common logged validation point within each panel, where lower is better; marker shape denotes within-step depth L , and stars mark the best configurations within each panel.



Async-paper Figure 2. FineWeb fixed-recurrent-compute allocation across three budgets. **Left:** small budget, with all 14 configurations complete. **Middle:** medium budget, twice the small-sweep recurrent compute, with all 16 configurations complete. **Right:** large budget, eight times the small-sweep recurrent compute, with 24 of 30 configurations complete, including completed low-expert and depth-16 extensions. All panels share one validation-loss color scale. Colored points and labels report the final validation checkpoint near 1.295B consumed tokens; all but one use W&B step 4,940, while the lower-PDB $D = 1, E = 2, W = 6144$ run uses its nearest checkpoint at step 4,953. Hollow points mark planned configurations that had not completed when this snapshot was generated. Circles, squares, triangles, filled-plus markers, and crosses denote $D = 1, 2, 4, 8, 16$, respectively; a star marks the best completed configuration in each panel. Lower loss is better. All runs use sequence length 128, memory length 128, a 262,144-token optimizer batch, sandwich RMSNorm, and a persistent K/V cache shared across depth. The large panel remains interim while its hollow configurations finish.

Table 10: Four-H100 median steady-state decode latency in milliseconds. GPT-16 uses cached autoregressive decoding with a full sliding-window key-value cache; PRE uses segment-local recurrent decoding with depth-shared recurrent key-value memory. B is the global batch and $B/4$ is the per-GPU batch.

Model	$C = 128, B = 16$	$C = 2048, B = 16$
GPT-16	0.636	0.760
PRE $D = 1, E = 16$	0.133	0.209
PRE $D = 2, E = 8$	0.498	0.645
PRE $D = 4, E = 4$	0.694	0.943
PRE $D = 8, E = 2$	0.943	1.447
PRE $D = 16, E = 1$	1.465	2.454

Table 11: Four-H100 FineWeb training throughput in tokens/s. Each panel fixes token batch CB and reports segment length C and global sequence batch B . Values are medians over steps 2–5; startup, compilation, and autotuning steps 0–1 are excluded. OOM denotes measured memory exhaustion. All displayed values use one microbatch per optimizer update.

131k effective tokens per optimizer update						
Model	$C = 32$ $B = 4,096$	$C = 64$ $B = 2,048$	$C = 128$ $B = 1,024$	$C = 256$ $B = 512$	$C = 1024$ $B = 128$	$C = 2048$ $B = 64$
GPT-16	1,004.7k	1,056.6k	1,098.2k	1,113.8k	1,066.1k	1,021.9k
PRE $D = 1, E = 16$	471.3k	417.4k	327.2k	218.7k	67.0k	OOM
PRE $D = 2, E = 8$	229.8k	182.0k	128.1k	79.2k	22.0k	OOM
PRE $D = 4, E = 4$	169.0k	134.6k	96.7k	60.6k	17.1k	OOM
PRE $D = 8, E = 2$	113.8k	88.5k	63.2k	39.4k	11.7k	OOM
PRE $D = 16, E = 1$	68.7k	54.1k	37.7k	24.3k	7.5k	OOM

262k effective tokens per optimizer update						
Model	$C = 32$ $B = 8,192$	$C = 64$ $B = 4,096$	$C = 128$ $B = 2,048$	$C = 256$ $B = 1,024$	$C = 1024$ $B = 256$	$C = 2048$ $B = 128$
GPT-16	1,115.5k	1,205.6k	1,238.8k	1,245.7k	1,168.6k	1,125.4k
PRE $D = 1, E = 16$	511.1k	486.0k	419.3k	319.4k	OOM	OOM
PRE $D = 2, E = 8$	263.7k	228.7k	179.4k	121.7k	OOM	OOM
PRE $D = 4, E = 4$	193.9k	166.6k	131.0k	90.0k	OOM	OOM
PRE $D = 8, E = 2$	128.9k	110.2k	84.8k	57.9k	OOM	OOM
PRE $D = 16, E = 1$	77.9k	66.3k	50.5k	34.1k	OOM	OOM

524k effective tokens per optimizer update						
Model	$C = 32$ $B = 16,384$	$C = 64$ $B = 8,192$	$C = 128$ $B = 4,096$	$C = 256$ $B = 2,048$	$C = 1024$ $B = 512$	$C = 2048$ $B = 256$
PRE $D = 1, E = 16$	543.6k	517.3k	477.9k	377.4k	OOM	OOM
PRE $D = 2, E = 8$	285.2k	260.2k	219.6k	157.1k	OOM	OOM
PRE $D = 4, E = 4$	208.2k	188.9k	158.9k	112.0k	OOM	OOM
PRE $D = 8, E = 2$	137.6k	124.0k	103.8k	OOM	OOM	OOM
PRE $D = 16, E = 1$	83.7k	74.4k	60.4k	OOM	OOM	OOM